

IB MATH AA SL

Lesson 6

Logarithms & Their Laws

Number & Algebra · 60-minute lesson

Syllabus SL 1.7 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. 2 to what power gives 8?
2. 10 to what power gives 1000?
3. Rewrite $3^4 = 81$ as a logarithm.

Answers (click to reveal)

1. 3 ($2^3 = 8$)
2. 3 ($10^3 = 1000$)
3. $\log_3 81 = 4$

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
a logarithm as the inverse of a power.
- 2 Convert**
between index and log form.
- 3 Apply**
the laws of logarithms.
- 4 Evaluate**
logs without a calculator where possible.

SUCCESS CRITERIA — I CAN...

- ✓
switch between $a^c = b$ and $\log_a b = c$
- ✓
use the product, quotient and power laws
- ✓
simplify expressions to a single logarithm
- ✓
evaluate simple logarithms exactly

What a logarithm is

DEFINITION

$$\square_a b = c \quad a^c = b$$

A log answers: 'what power gives this number?'

$\log_2 8 = 3$ because $2^3 = 8$.

KEY VALUES

$$\log_a 1 = 0$$

$$a^0 = 1$$

$$\log_a a = 1$$

$$a^1 = a$$

log means $\log_{10}$

ln means $\log_e$

Evaluate a log

QUESTION

Evaluate $\log_2 32$ without a calculator.

SOLUTION

Step 1 ask: 2 to what power is 32?

Step 2 write 32 as a power of 2:

$$32 = 2^5$$

Step 3 so the logarithm is the index:

$$\log_2 32 = 5$$

Your turn — evaluate

ANSWER IN YOUR BOOK

1. $\log_3 9$
2. $\log_5 125$
3. $\log_{10} 1000$

Answers (click to reveal)

1. 2
2. 3
3. 3

The laws of logs

THE LAWS

$$\log_a xy = \log_a x + \log_a y$$

$$\log_a \frac{x}{y} = \log_a x - \log_a y$$

$$\log_a x^n = n \log_a x$$

Product

→ add the logs

Quotient

→ subtract

Power

USES

Combine several logs into one.

Split one log to solve equations.

Same base

the laws only work with a common base.

Use the laws

QUESTION

Write as a single logarithm:

$$\log_3 9 + \log_3 27.$$

SOLUTION

Step 1 evaluate each (or use the product law):

$$\log_3 9 = 2, \log_3 27 = 3$$

Step 2 add them:

$$2 + 3 = 5$$

Step 3 (check: $\log_3(9 \times 27) = \log_3 243 = 5$)

$$= 5$$

Exam-style question

- (a) Write $\log x + 2 \log y$ as a single logarithm. [2]
(b) Given $\log_2 5 = k$, write $\log_2 20$ in terms of k . [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $\log x + \log y^2 = \log(xy^2)$ (M1)(A1)
(b) $\log_2 20 = \log_2(4 \times 5) = \log_2 4 + \log_2 5$ (M1) $= 2 + k$ (A1)(A1)

Common mistakes



log of a sum

$\log(x+y) \neq \log x + \log y$ — the product law is for $\log(xy)$.



Base matters

You can only combine logs with the same base.



Power law

$\log x^n = n \log x$ — the index comes to the front.



$\log_a 1 = 0$

Not 1 — any base to the power 0 is 1.

Mixed practice

QUESTIONS

1. Evaluate $\log_4 64$
2. Write $\log 8 - \log 2$ as one log
3. Simplify $3 \log x - \log y$
4. Evaluate $\log_2 1$

Answers (click to reveal)

1. 3
2. $\log 4$
3. $\log(x^3/y)$
4. 0

Plenary & exit ticket

KEY FORMULAS

$$\square_a b = c \quad a^c = b$$

$$\square_a x y = \square_a x + \square_a y$$

EXIT TICKET — before you leave

1. Evaluate $\log_3 81$
2. Write $\log 5 + \log 4$ as one log
3. Simplify $\log_2 2^5$

Answers (click to reveal)

1. 4
2. $\log 20$
3. 5

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Laws of logarithms.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 7

Solving exponential & logarithmic equations using these laws.

$$a^x = b \Rightarrow x = \square_a b$$